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**UNIVERSITY OF CAMBRIDGE LOCAL EXAMINATIONS SYNDICATE**

**Joint Examination for the School Certification and General Certificate of Education Ordinary Level**

# CHEMISTRY 5070/1

PAPER 1 Multiple Choice

**OCTOBER/NOVEMBER SESSION 2002**

1 hour

Additional materials:
- Multiple Choice answer sheet
- Soft clean eraser
- Soft pencil (type B or HB is recommended)

**TIME** 1 hour

## INSTRUCTIONS TO CANDIDATES

**Do not open this booklet until you are told to do so.**

Write your name, Centre number and candidate number on the answer sheet in the spaces provided unless this has already been done for you.

There are **forty** questions in this paper. Answer **all** questions. For each question, there are four possible answers, **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in **soft pencil** on the separate answer sheet.

**Read very carefully the instructions on the answer sheet.**

## INFORMATION FOR CANDIDATES

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

A copy of the Periodic Table is printed on page 16.

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**1** Which property of a gas affects the rate at which it spreads throughout a laboratory?

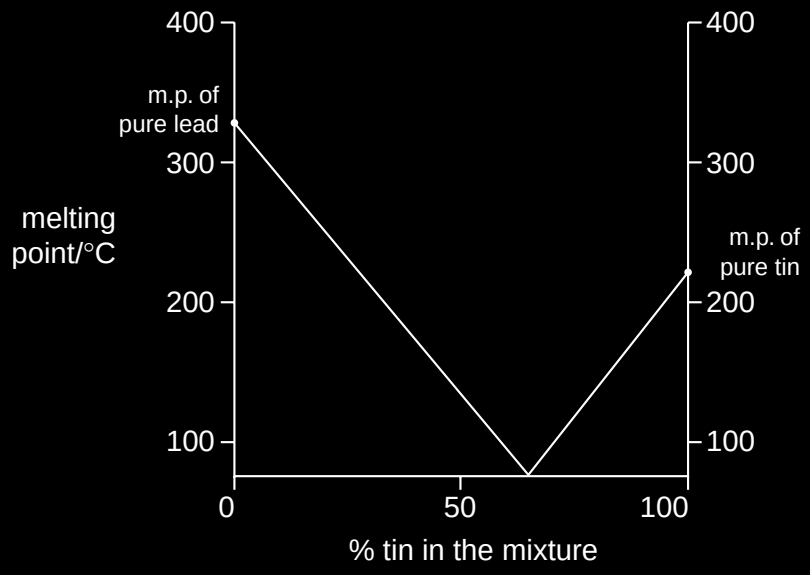
**A** boiling point

**B** molecular mass

**C** reactivity

**D** solubility in water

**2** The graph gives the melting points of mixtures of lead and tin.




The graph shows that any mixture of lead and tin must have a melting point

**A** above that of tin.

**B** below that of lead.

**C** below that of both tin and lead.

**D** between that of tin and lead.

**3** From which mixture can the underlined substance be obtained by adding water, stirring and filtering?

**A** <u>calcium carbonate</u> and sodium chloride

**B** copper(II) sulphate and <u>sodium chloride</u>

**C** <u>ethanoic acid</u> and ethanol

**D** iron and <u>magnesium</u>

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**4** An aqueous solution of a sulphate is made from a solid hydroxide, of a metal **M**, by th

$$M(OH)_2\,(s) + H_2SO_4(aq) \longrightarrow MSO_4(aq) + 2H_2O(l)$$

For which hydroxide would the method **not** work?

**A** barium hydroxide

**B** copper(II) hydroxide

**C** iron(II) hydroxide

**D** magnesium hydroxide

**5** Which ion has the most shells that contain electrons?

**A** $Al^{3+}$

**B** $Be^{2+}$

**C** $N^{3-}$

**D** $S^{2-}$

**6** The table gives data about four substances.

Which substance could be an ionic compound?

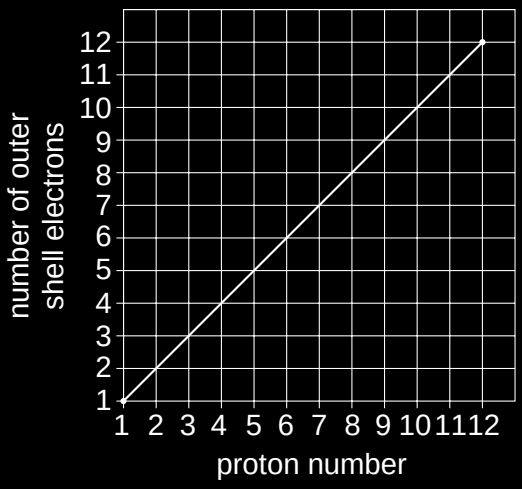
| compound | melting point / °C | electrical conductivity in aqueous solution |
|---|---|---|
| **A** | –73 | good |
| **B** | 32 | poor |
| **C** | 474 | poor |
| **D** | 805 | good |

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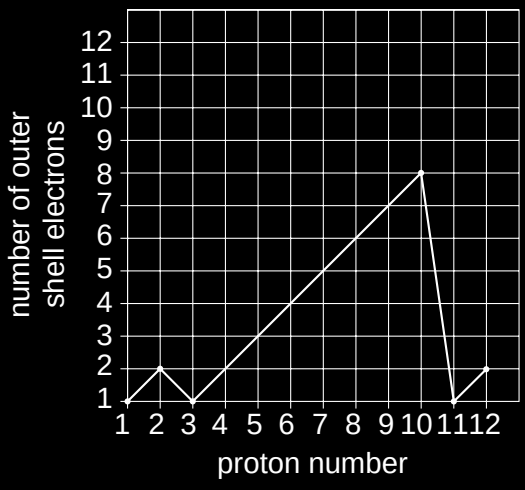
**7** The number of outer shell electrons for the atoms of the first 12 elements in the Periodic Table is plotted against the proton number of the element.

Which graph is obtained?

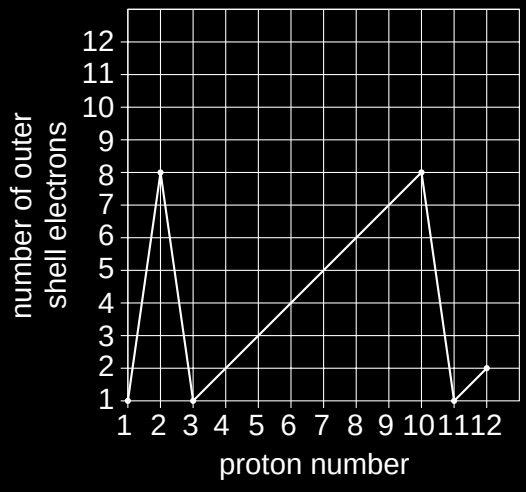
**A**



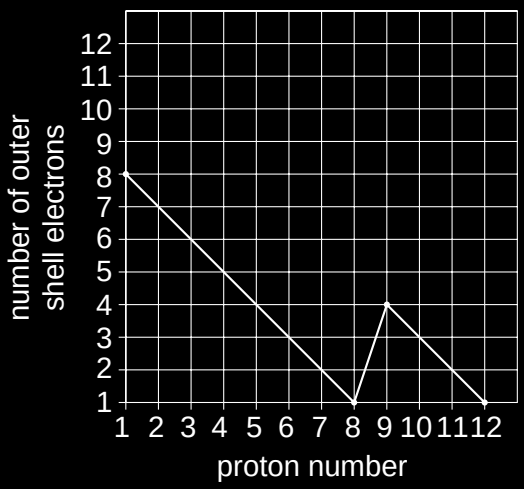

**B**




**C**




**D**



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**8** The table shows the electron structures of four elements.

| element | electronic structure |
|---|---|
| **W** | 2, 6 |
| **X** | 2, 8 |
| **Y** | 2, 8, 1 |
| **Z** | 2, 8, 7 |

Which pair of atoms will form a covalent substance?

**A** two atoms of **W**

**B** two atoms of **X**

**C** an atom of **W** and an atom of **X**

**D** an atom of **Y** and an atom of **Z**

**9** Which substance contains covalent bonds, but also conducts electricity?

**A** brass

**B** graphite

**C** iodine

**D** steel

**10** One mole of each of the following compounds is burnt in excess oxygen.

Which compound will produce three moles of carbon dioxide and three moles of steam only?

**A** $C_3H_8$ **B** $C_3H_7OH$ **C** $C_3H_7CO_2H$ **D** $CH_3CO_2CH_3$

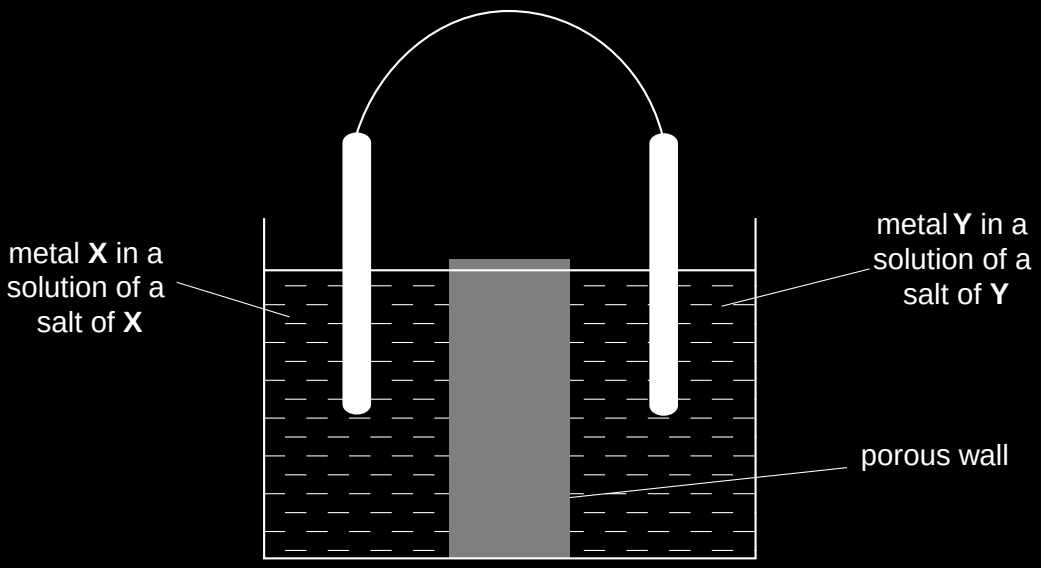
**11** When zinc reacts with dilute sulphuric acid a gas is released.

What happens to the zinc and what is the gas released?

| | the zinc is | the gas is |
|---|---|---|
| **A** | oxidised | hydrogen |
| **B** | oxidised | sulphur dioxide |
| **C** | reduced | hydrogen |
| **D** | reduced | sulphur dioxide |

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**12** Which pair of metals **X** and **Y** will produce the highest voltage when used as ele[illegible] simple cell?




| | metal **X** | metal **Y** |
|---|---|---|
| **A** | copper | silver |
| **B** | magnesium | silver |
| **C** | magnesium | zinc |
| **D** | zinc | copper |

**13** Four electrolytes were electrolysed using carbon electrodes.

Which set of data is correct?

| | electrolyte | product at | |
|---|---|---|---|
| | | anode | cathode |
| **A** | $CuSO_4$ (aq) | oxygen | copper |
| **B** | $NaCl$ (aq) | chlorine | sodium |
| **C** | NaH (l) | sodium | hydrogen |
| **D** | $PbBr_2$ (l) | lead | bromine |

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**14** Which pair of substances are isotopes?

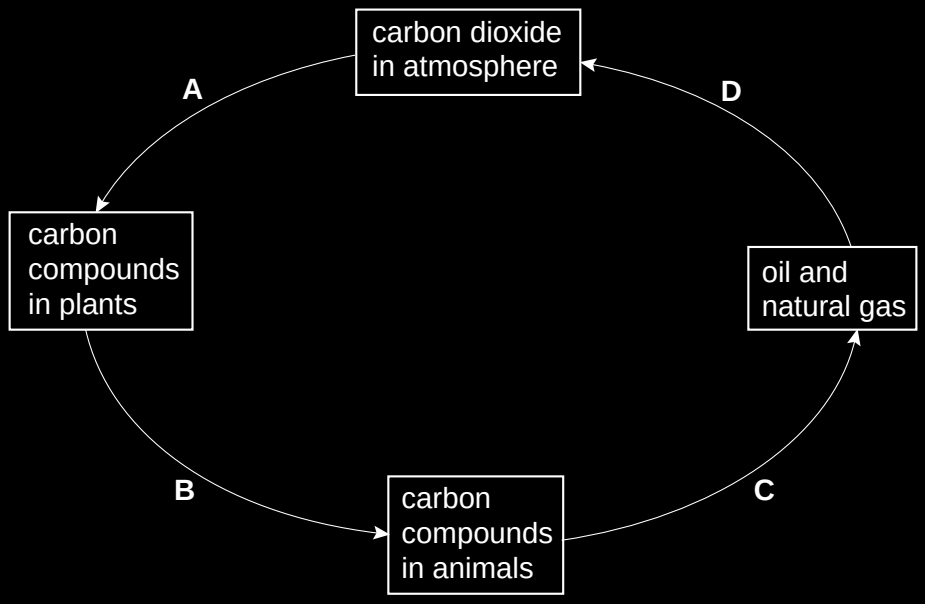
**A** $^{12}_{6}C$ and $^{14}_{6}C$

**B** carbon dioxide and carbon monoxide

**C** diamond and graphite

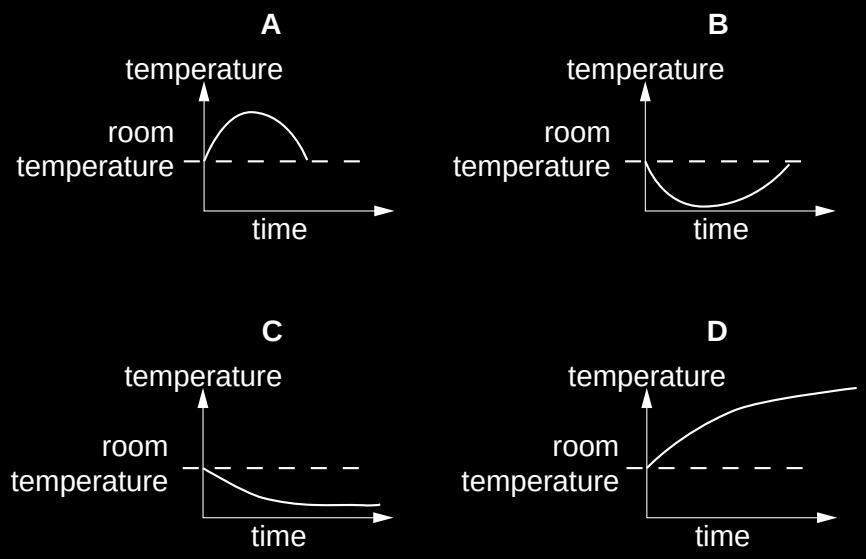
**D** $C_2H_4$ and $C_3H_6$

**15** Which step in the diagram shows the process of photosynthesis?




**16** Dissolving ammonium nitrate in water is endothermic.

Which graph shows how the temperature alters as the ammonium nitrate is added to water and then the solution is left to stand?



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**17** If a strip of magnesium is dropped into excess hydrochloric acid an exothermic reacti

The rate of this reaction increases during the first few seconds because

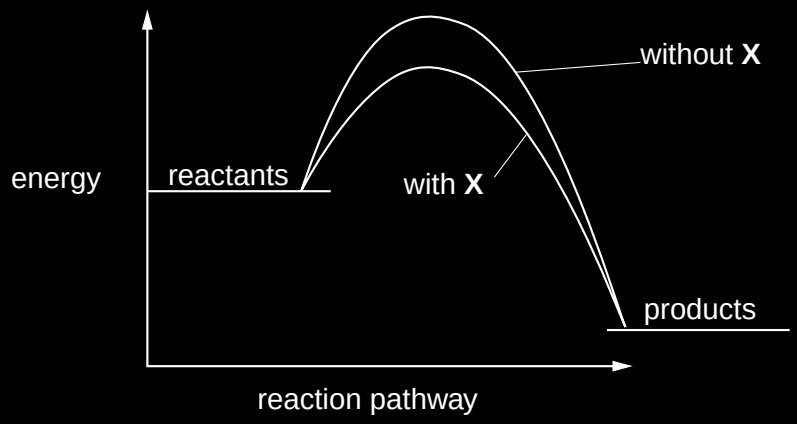
**A** the amount of magnesium is decreasing.

**B** the magnesium is acting as a catalyst.

**C** the solution is becoming hotter.

**D** the surface area of the magnesium is increasing.

**18** The energy profile diagrams show how adding a substance **X** to a reaction mixture changes the reaction pathway.




Which change is likely to be observed when **X** is added to the reaction mixture?

**A** The reaction becomes less exothermic.

**B** The reaction becomes more exothermic.

**C** The speed of the reaction decreases.

**D** The speed of the reaction increases.

**19** Which process does **not** involve either oxidation or reduction?

**A** formation of ammonium sulphate from ammonia and sulphuric acid

**B** formation of nitrogen monoxide from ammonia

**C** formation of sulphuric acid from sulphur

**D** formation of zinc from zinc blende (ZnS)

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**20** In separate experiments, an excess of aqueous sodium hydroxide or aqueous ammonia is gradually added to a solution **X**.

In both experiments, a precipitate was obtained which dissolved in an excess of the alkali reagent.

What could **X** contain?

**A** copper(II) nitrate

**B** iron(II) nitrate

**C** iron(III) nitrate

**D** zinc nitrate

**21** An excess of dilute sulphuric acid reacts with both aqueous barium hydroxide and aqueous barium chloride. In what way are the two reactions the same?

**A** A gas is produced.

**B** An insoluble salt is produced.

**C** The final pH is 7.

**D** Water is produced.

**22** Which property decides the order of the elements in the Periodic Table?

**A** the masses of their atoms

**B** the number of electrons in the outer shell

**C** the number of neutrons in the nucleus

**D** the number of protons in the nucleus

**23** The proton number of indium, In, is 49.

What is the most likely formula for the oxide of indium?

**A** $In_2O$ **B** $In_2O_3$ **C** $InO$ **D** $InO_2$

**24** Which element in the table is likely to be a transition metal?

| | *element* | *melting point* | *colour of chloride* |
|---|---|---|---|
| **A** | | high | blue |
| **B** | | low | green |
| **C** | | high | white |
| **D** | | low | white |

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**25** Which feature of a metal's structure is responsible for it conducting electricity?

**A** It contains positive ions.

**B** It has a "sea of electrons".

**C** Its ions are tightly packed together.

**D** Its positive ions attract electrons.

**26** Aluminium is extracted from purified bauxite by electrolysis but iron is extracted from haematite by reduction with coke.

Why is iron not extracted by electrolysis?

**A** Haematite needs to be purified but bauxite does not.

**B** Iron is less reactive than aluminium.

**C** Reduction with coke is cheaper than electrolysis.

**D** Reduction with coke gives a purer product than electrolysis.

**27** Old steel drums corrode quickly in a damp atmosphere but aluminium cans do not.

Which of the following correct statements explains this behaviour of aluminium?

**A** Aluminium has only one valency.

**B** Aluminium has a lower density than iron.

**C** Aluminium is above iron in the activity series.

**D** Aluminium is protected by its oxide layer.

**28** Caesium is a metal that is more reactive than aluminium.

Which reaction would produce caesium?

**A** electrolysing aqueous caesium chloride

**B** electrolysing molten caesium chloride

**C** heating caesium carbonate

**D** heating caesium oxide with carbon

**29** Which of the following gases **cannot** be removed from the exhaust gases of a petrol powered car by its catalytic converter?

**A** carbon dioxide

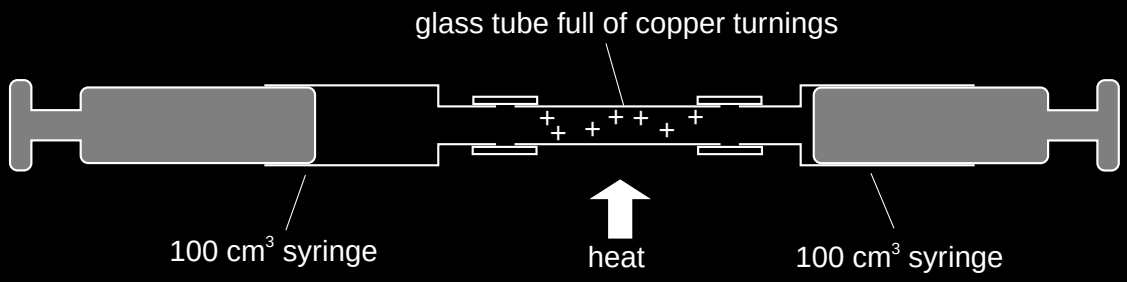
**B** carbon monoxide

**C** hydrocarbons

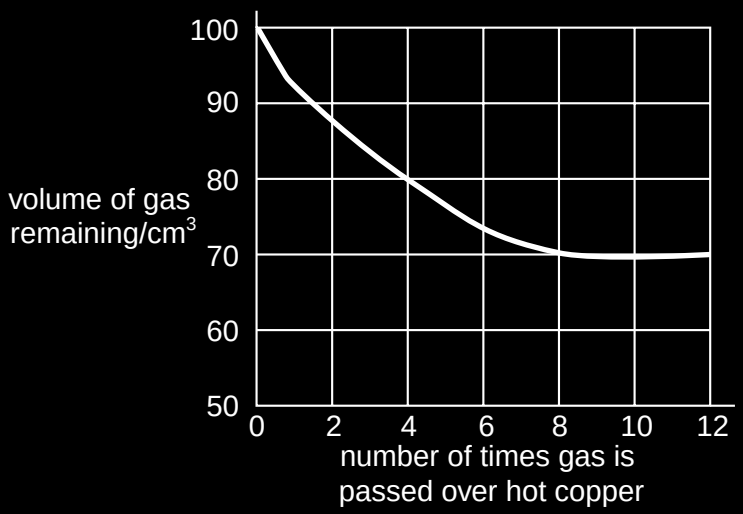
**D** nitrogen dioxide

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**30** A 100 $cm^3$ sample of bottled gas used for diving was placed in a gas syringe in th
shown.




The gas was passed backward and forward over heated copper turnings. The results obtained were used to plot the graph.




What is the percentage of oxygen in the bottled gas?

**A** 20% **B** 30% **C** 70% **D** 80%

**31** In the Haber process, nitrogen and hydrogen react to form ammonia.

$$N_2(g) + 3H_2(g) \rightleftharpoons 2\,NH_3(g) \quad \Delta H = -92\,kJ/mol$$

Which factor increases **both** the speed of reaction and the amount of ammonia produced?

**A** addition of a catalyst

**B** decreasing the temperature

**C** increasing the pressure

**D** increasing the temperature

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**32** Nitrates from fertilisers used on farmland can cause pollution.

Why do nitrates pollute rivers?

**A** Nitrates are salts.

**B** Nitrates are very soluble in water.

**C** Nitrates contain oxygen.

**D** Nitrate ions are negatively charged.

**33** Which representation of dilute sulphuric acid is correct?

**A** $H_2(aq) + SO_4^{2-}(aq)$

**B** $2H^+(aq) + SO_4^{2-}(aq)$

**C** $2H^+(aq) + SO_4^-(aq)$

**D** $H_2SO_4(l)$

**34** Which statement describes what happens when hydrogen and oxygen are used in a fuel cell?

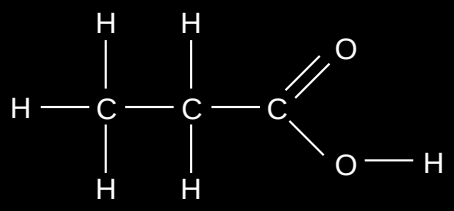
**A** Electricity is generated directly.

**B** Electricity is used to produce water.

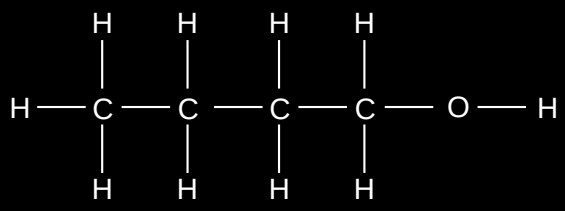
**C** Hydrogen is burned to form steam.

**D** Hydrogen reacts to form a hydrocarbon fuel.

**35** The structures of an acid and an alcohol are shown.



acid



alcohol

Which pairing of names correctly identify the two compounds?

| | acid | alcohol |
|---|---|---|
| **A** | ethanoic | butanol |
| **B** | ethanoic | propanol |
| **C** | propanoic | propanol |
| **D** | propanoic | butanol |

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**36** Which physical property of the alkanes does **not** increase as relative molecular mass

**A** boiling point

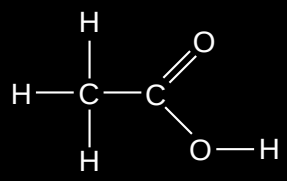
**B** flammability

**C** melting point

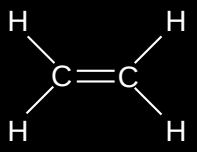
**D** viscosity

**37** The structures of four organic compounds are shown.

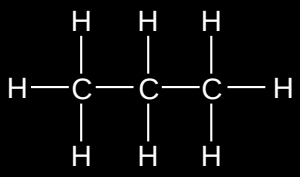
1



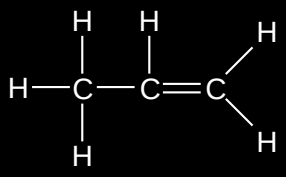
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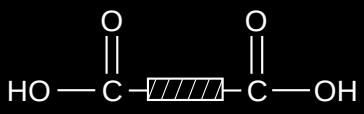
4



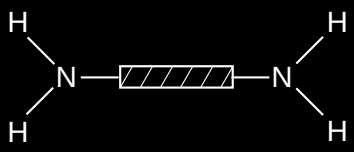
Which compounds decolourise bromine water?

**A** 1 and 2 **B** 1, 2 and 4 **C** 2 and 4 **D** 3 and 4

**38** A polymer X was hydrolysed and the two products were



and



What can be deduced about X?

**A** It was a condensation polymer.

**B** It was starch.

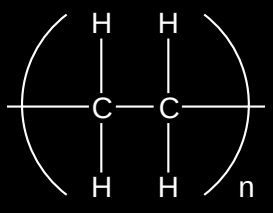
**C** It was made by addition polymerisation.

**D** It was *Terylene*.

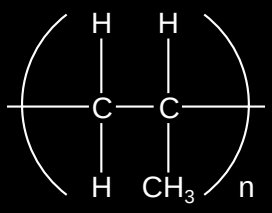
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**39** Which polymer has the empirical formula CH?

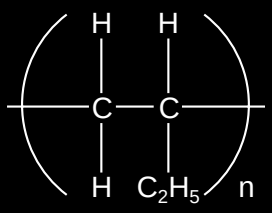
**A**



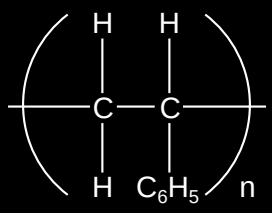

**B**




**C**




**D**




**40** In the polymerisation of ethene to form poly(ethene), there is no change in

**A** boiling point.

**B** density.

**C** mass.

**D** molecular formula.

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# DATA SHEET
## The Periodic Table of the Elements

| Group | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| I | II | | | | | | | | | | | III | IV | V | VI | VII | 0 |
| | | | | | | | 1 H Hydrogen 1 | | | | | | | | | | 4 He Helium 2 |
| 7 Li Lithium | 9 Be Beryllium 4 | | | | | | | | | | | 11 B Boron 5 | 12 C Carbon 6 | 14 N Nitrogen 7 | 16 O Oxygen 8 | 19 F Fluorine 9 | 20 Ne Neon 10 |
| 23 Na Sodium | 24 Mg Magnesium 12 | | | | | | | | | | | 27 Al Aluminium 13 | 28 Si Silicon 14 | 31 P Phosphorus 15 | 32 S Sulphur 16 | 35.5 Cl Chlorine 17 | 40 Ar Argon 18 |
| 39 K otassium | 40 Ca Calcium 20 | 45 Sc Scandium 21 | 48 Ti Titanium 22 | 51 V Vanadium 23 | 52 Cr Chromium 24 | 55 Mn Manganese 25 | 56 Fe Iron 26 | 59 Co Cobalt 27 | 59 Ni Nickel 28 | 64 Cu Copper 29 | 65 Zn Zinc 30 | 70 Ga Gallium 31 | 73 Ge Germanium 32 | 75 As Arsenic 33 | 79 Se Selenium 34 | 80 Br Bromine 35 | 84 Kr Krypton 36 |
| 85 Rb Rubidium | 88 Sr Strontium 38 | 89 Y Yttrium 39 | 91 Zr Zirconium 40 | 93 Nb Niobium 41 | 96 Mo Molybdenum 42 | Tc Technetium 43 | 101 Ru Ruthenium 44 | 103 Rh Rhodium 45 | 106 Pd Palladium 46 | 108 Ag Silver 47 | 112 Cd Cadmium 48 | 115 In Indium 49 | 119 Sn Tin 50 | 122 Sb Antimony 51 | 128 Te Tellurium 52 | 127 I Iodine 53 | 131 Xe Xenon 54 |
| 133 Cs Caesium | 137 Ba Barium 56 | 139 La Lanthanum 57 * | 178 Hf Hafnium 72 | 181 Ta Tantalum 73 | 184 W Tungsten 74 | 186 Re Rhenium 75 | 190 Os Osmium 76 | 192 Ir Iridium 77 | 195 Pt Platinum 78 | 197 Au Gold 79 | 201 Hg Mercury 80 | 204 Tl Thallium 81 | 207 Pb Lead 82 | 209 Bi Bismuth 83 | Po Polonium 84 | At Astatine 85 | Rn Radon 86 |
| Fr Francium | 226 Ra Radium 88 | 227 Ac Actinium 89 † | | | | | | | | | | | | | | | |

| | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 140 Ce Cerium 58 | 141 Pr Praseodymium 59 | 144 Nd Neodymium 60 | Pm Promethium 61 | 150 Sm Samarium 62 | 152 Eu Europium 63 | 157 Gd Gadolinium 64 | 159 Tb Terbium 65 | 162 Dy Dysprosium 66 | 165 Ho Holmium 67 | 167 Er Erbium 68 | 169 Tm Thulium 69 | 173 Yb Ytterbium 70 | 175 Lu Lutetium 71 |
| 232 Th Thorium 90 | Pa Protactinium 91 | 238 U Uranium 92 | Np Neptunium 93 | Pu Plutonium 94 | Am Americium 95 | Cm Curium 96 | Bk Berkelium 97 | Cf Californium 98 | Es Einsteinium 99 | Fm Fermium 100 | Md Mendelevium 101 | No Nobelium 102 | Lr Lawr 10 |

-71 Lanthanoid series
0-103 Actinoid series

y | a X b | a = relative atomic mass; X = atomic symbol; b = proton (atomic) number

The volume of one mole of any gas is 24 $dm^3$ at room temperature and pressure (r.t.p.).

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